

SMART INDIA HACKATHON 2026

National Idea Presentation · Problem Statement ID: SIH26062

OFFICIAL REGISTRATION DETAILS

Problem Statement ID: SIH26062

Problem Statement Title: Integrated Polar Expedition Logistics And Asset Management System

Theme: Smart Automation

PS Category: Software

Team Name: CompileX (Registered on Portal)

Team ID: - (As recorded on portal)

PROJECT IDENTITY & POSITIONING

POLAR-AI

Polar Operations, Logistics & Autonomous Resilience Intelligence

CORE MISSION STATEMENT:

Mission Continuity Intelligence for Extreme Environments

POLAR-AI transforms fragmented manifests, sensor logs, and operational records into proactive intelligence — monitoring resource runways, predicting cascading multi-tier outages, and enabling safe human decisions under extreme isolation.

Dual-Theatre Architecture: Antarctica (Maitri & Bharati) · Arctic (Himadri, Svalbard) · Logistics (NCPOR Goa)

Predictive Continuity

Dynamic 0–100% telemetry scoring & shortage horizon alerts

What-If Simulation

Models supply delays, consumption surges, & ration trade-offs

100% Offline Autonomy

Local PWA storage & deterministic conflict-free sync on reconnect

FROM FRAGMENTED MISSION DATA TO MISSION CONTINUITY INTELLIGENCE

Bridging the critical operational gap between passive logistics tracking and proactive mission survival.

THE POLAR CHALLENGE: WHY PASSIVE TRACKING FAILS

1. Fragmented Data Silos

Cargo manifests, stock levels, equipment health, and rosters remain isolated across manual spreadsheets.

2. Resupply Uncertainty

8-month winter freeze-out; blizzards and pack ice cause shipping delays where consumption surges create fatal deficits.

3. Limited Operational Visibility

Station commanders lack a unified cockpit to correlate incoming cargo ETA against critical station burn rates.

4. Cascading Dependencies

Failure of one resource cascades: Fuel Deficit → Generator G-01 Trip → Station Power Loss → Heating Failure → Lab Halted.

5. Connectivity Constraints

Severe polar blizzards & geomagnetic interference break satellite links; field sorties operate in complete communication silence.

**"Knowing what changed is not enough.
Operators need to understand what that change means for mission continuity."**

POLAR-AI SOLUTION: END-TO-END RESILIENCE PIPELINE

STAGE 1: UNIFIED MISSION DATA INGESTION

Integrates Cargo manifests, Inventory burn-rates, Machinery wear, Personnel, & Weather into a single relational state model.

STAGE 2: MISSION CONTINUITY ENGINE

Computes composite 0–100% continuity score, calculates days-to-stockout runway, & evaluates directed dependency graph linkages.

STAGE 3: PREDICTIVE & CASCADING IMPACT ANALYSIS

Detects emerging risks 7–14 days ahead; explains root causes and traces multi-tier asset cascade risks automatically.

STAGE 4: WHAT-IF SIMULATION & MITIGATION OPTIONS

Allows operators to test resupply delays, fuel rationing, or generator load-shedding before real-world operational execution.

STAGE 5: SOVEREIGN HUMAN DECISION SUPPORT

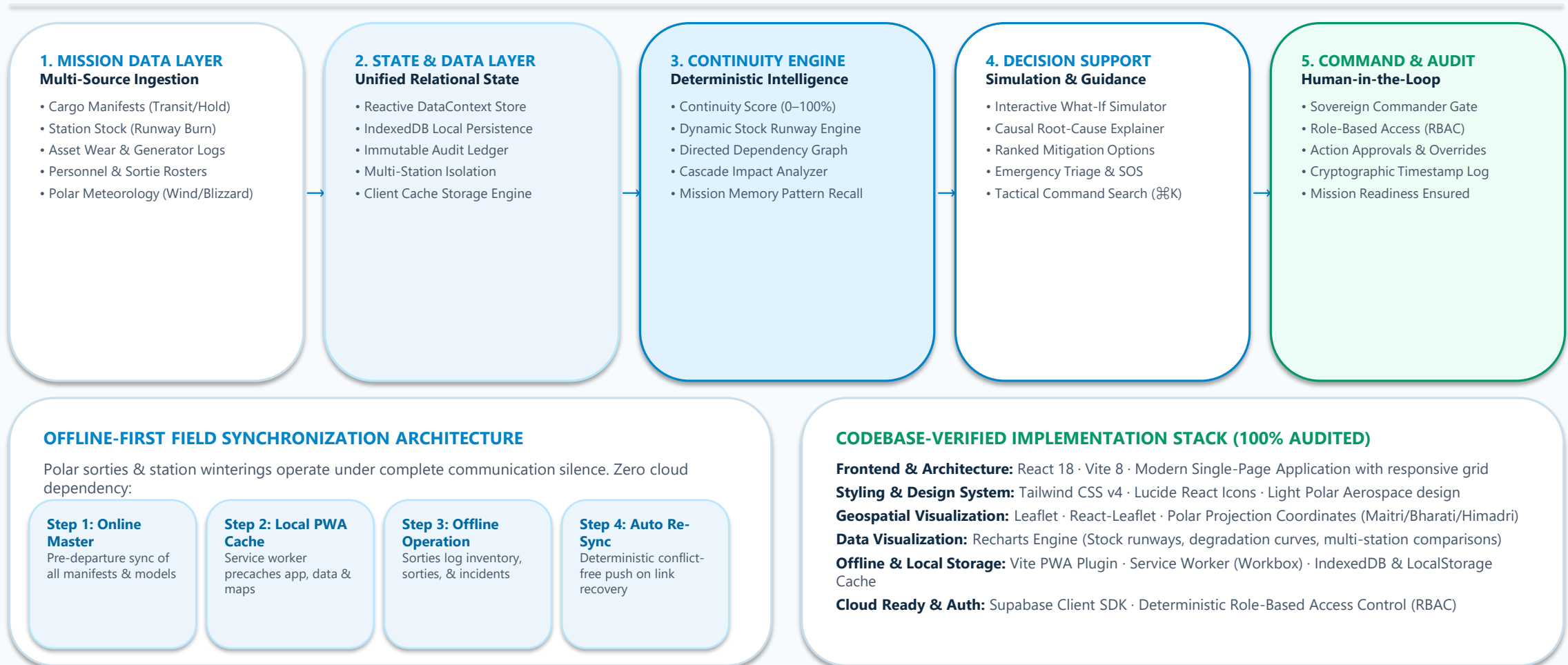
Synthesizes ranked operational mitigations; keeps sovereign human commander in absolute control (Review → Approve/Modify → Audit).

CORE USP — MISSION CONTINUITY INTELLIGENCE

"POLAR-AI connects mission resources, dependencies and operational risks to predict cascading impacts and test mitigation options before they become mission-critical."

TECHNICAL APPROACH: ZERO-BANDWIDTH RESILIENT ARCHITECTURE

Diagram-first end-to-end mission intelligence flow with local-first persistence and zero cloud dependency.



FEASIBILITY & VIABILITY: RIGOROUS EXTREME-ENVIRONMENT FIT

Demonstrating working software capabilities, operational constraints handling, and scalable deployment.

1. TECHNICAL FEASIBILITY (DEMONSTRATED IN CODEBASE)

- **Mission Continuity Engine:** Live 0–100% composite health index recalculating across real operational data.
- **What-If Simulator:** Interactive perturbation of resupply delays, consumption surges, & generator shutdowns.
- **Cascading Dependency Graph:** Visual linkage of assets (Fuel → Generators → Power → Heating → Science).
- **Offline-First PWA:** Service worker precaching assets; full navigation & logging without active internet.
- **Mission Memory Intelligence:** Synthesizes lessons from historical polar logs to advise active sorties.

2. OPERATIONAL FEASIBILITY (POLAR ENVIRONMENT FIT)

- **Remote Polar Deployment:** Built for Antarctic stations (Maitri, Bharati) & Arctic research (Himadri).
- **Low-Bandwidth Hardware Tolerance:** Ultra-compact web bundle (<1.2MB gzip) runs on low-power field rugged laptops.
- **Strict Resource Constraints:** Models non-negotiable fuel, water, ration, & medical buffer reserve limits.
- **Centralized Mission Command:** Unifies logistics, science, and medical officers under synchronized theatre state.
- **Human-in-the-Loop Governance:** Zero autonomous overrides of life-support; decisions require commander signature.

3. OPERATIONAL CHALLENGES & VERIFIED MITIGATIONS

- **Limited / Zero Connectivity** → Offline-first PWA caching with persistent IndexedDB & local queue.
- **Concurrent Offline Data Edits** → Deterministic timestamped state reconciliation upon link restoration.
- **Complex Multi-System Dependencies** → Directed dependency graphs tracing ripple effects before decisions.
- **Untrusted AI Recommendations** → Strict Human-in-the-Loop gate; explainable rule heuristics & audit logs.
- **Dynamic Extreme Weather** → What-If simulation recalculates burn & runway prior to sortie dispatch.

4. STRATEGIC SCALABILITY ROADMAP

- **Phase 1: Indian Polar Expeditions:** Deployment across Antarctic stations (Maitri, Bharati) & Arctic base (Himadri).
- **Phase 2: Multi-Station Network:** Synchronization across supply vessels (MV Vasily Golovnin), aircraft, & field camps.
- **Phase 3: High-Altitude Defense & Science:** Extensible to Siachen, Ladakh, and Himalayan glaciological outposts.
- **Phase 4: Disaster Relief Operations:** Blackout logistics coordination for island emergencies (Andaman) & flood relief.
- **Phase 5: Extreme Industrial Sectors:** Offshore oil platforms, deep-sea exploration, and remote mining sites.

IMPACT & BENEFITS: PROTECTING POLAR SCIENCE & EXPEDITION LIVES

Delivering operational clarity, early hazard awareness, and defensible decision support for national polar programs.

1. UNIFIED OPERATIONAL VISIBILITY

- **Single Glass Cockpit:** Consolidates distributed manifests, stock ledgers, and machinery telemetry into one real-time operational picture.
- **Complete Cargo Pipeline:** Tracks multi-stage freight movement from NCPOR Goa through Cape Town to Antarctic ice shelf offloading.
- **Real-Time Runway Calculations:** Replaces guesswork with dynamic burn-rate projections for fuel, water, medical supplies, and food.
- **Multi-Station Oversight:** Allows expedition commanders to monitor Maitri, Bharati, and Himadri stations simultaneously.

2. EARLY RISK AWARENESS & CASCADE DETECTION

- **7–14 Day Predictive Horizon:** Detects impending inventory breaches before emergency reserves are depleted.
- **Dependency Cascade Tracing:** Identifies hidden secondary impacts (e.g. generator fuel starvation causing lab freeze-outs).
- **Weather & Sea-Ice Correlation:** Flags vessel approach hazards and blizzard windows that impact surface traverse safety.
- **Root-Cause Attribution:** Explains exactly why an alert fired rather than outputting cryptic anomaly scores.

3. DEFENSIBLE DECISION SUPPORT & WHAT-IF SIMULATION

- **Safe Synthetic Experimentation:** Operators test fuel rationing or load shedding in simulation prior to real-world deployment.
- **Ranked Mitigation Trade-Offs:** AI recommends prioritized operational responses with explicit runway deltas.
- **Sovereign Human Command Gate:** Preserves commander authority; no action is taken without explicit authorized review.
- **Immutable Audit Traceability:** Every decision, justification, and operator ID is cryptographically timestamped.

4. MISSION RESILIENCE & RESOURCE EFFICIENCY

- **Prevention of Crisis Evacuations:** Early mitigation avoids life-threatening mid-winter polar rescues and emergency air drops.
- **Elimination of Duplicate Orders:** Centralized cargo tracking prevents unnecessary intercontinental supply shipments.
- **Environmental Treaty Compliance:** Aids Antarctic Treaty waste management and hazmat tracking protocols.
- **Continuous Field Readiness:** Ensures research teams maintain unbroken power and life-support across the 240-day polar night.

SCALABILITY BEYOND POLAR EXPEDITIONS: Extreme High-Altitude Border Logistics (Siachen/Ladakh) · Disaster Relief Operations (NDRF) · Offshore Energy Platforms · Island Emergency Supply Chains

RESEARCH & OPERATIONAL REFERENCES

Rooted in authentic Indian polar expedition mandates, international treaty frameworks, and empirical field logistics.

1. DOMAIN & INSTITUTIONAL CONTEXT

NCPOR Mandate:

National Centre for Polar and Ocean Research (MoES, Govt. of India) coordinates scientific research and multi-modal logistics across polar sectors.

Indian Research Bases:

Antarctica: Maitri (1989, Schirmacher Oasis) & Bharati (2012, Larsemann Hills).

Arctic: Himadri (2008, Ny-Ålesund, Svalbard, 78°55' N).

Antarctic Treaty System (ATS):

Strict governance under Madrid Protocol (1991) regulating waste repatriation, fuel bunkering, and zero environmental contamination.

COMNAP Standards:

Council of Managers of National Antarctic Programs operational guidelines for inter-station SAR and cold-region safety.

2. POLAR OPERATIONAL INSIGHTS

Multi-Stage Resupply Corridor:

Cargo moves across complex maritime pipelines: NCPOR Goa → Mumbai Port → Cape Town Gateway → Antarctic Ice Shelf → Continental Traverse.

The Winter Isolation Window:

Stations face complete physical isolation for ~8 months (March–November); sea ice blocks vessels and storms ground flights.

Critical Power & Heating Spine:

Station survival depends on uninterrupted Arctic-grade Jet A-1 / ATF fuel for combined heat & power (CHP) generators.

Environmental Retrograde:

All station solid waste, sewage sludge, and used lubricants must be manifested, compacted, and shipped back to the mainland.

3. CITATIONS & PROJECT REPOSITORY

NCPOR Official Portal & Expedition Logs:

National Centre for Polar and Ocean Research, Ministry of Earth Sciences, Govt. of India (ncpor.res.in).

Ministry of Earth Sciences (MoES):

Annual Reports & Indian Antarctic Research Operational Guidelines (moes.gov.in).

Secretariat of the Antarctic Treaty:

Protocol on Environmental Protection to the Antarctic Treaty — Annex III: Waste Disposal & Waste Management (ats.aq).

COMNAP Operations Manual:

Council of Managers of National Antarctic Programs — Station Logistics & Flight Safety Directives (comnap.aq).

POLAR-AI Software Prototype:

Smart India Hackathon 2026 Submission Repository:

github.com/bhavishyaone1/POLAR-AI

Verified React 18, Vite 8, PWA & Leaflet implementation.